

Milestone-3 (Geographical Analysis of US Disaster Declarations)

1. Introduction

Natural disasters occur across different regions of the United States, but their frequency and type vary depending on geographic conditions. Coastal states often experience hurricanes and storms, while western states are more prone to wildfires and earthquakes.

The purpose of this milestone is to perform geographical analysis of disaster declarations and identify regional patterns and disaster hotspots across the United States.

By visualizing disaster occurrences on maps and comparative charts, spatial patterns become easier to understand than when viewing raw tabular data.

This milestone focuses on transforming disaster data into state-level geographic insights using data aggregation and visualization techniques.

2. Dataset Description

The dataset used in this milestone contains historical records of disaster declarations issued in the United States.

Each record represents a disaster event declared in a particular state along with its disaster type and time information.

The dataset used for analysis is the cleaned dataset produced in the previous milestone.

Dataset File

data/processed/usnd_cleaned.csv

Columns Used

Column	Description
state	Two-letter abbreviation representing the US state
incidentType	Type of disaster (Flood, Hurricane, Fire, etc.)
year	Year of disaster declaration
month	Month of disaster declaration

These columns allow analysis of where disasters occur and what types of disasters dominate in different locations.

3. Tools and Libraries Used

The analysis was implemented using Python and several data science libraries.

Data Processing

➤ **Pandas**

Used for:

- Reading datasets
- Cleaning and transforming data
- Aggregating disaster counts

➤ **NumPy**

Used for numerical operations and data manipulation.

Visualization Libraries

➤ **Matplotlib**

Used to generate basic charts such as bar charts.

➤ **Seaborn**

Used for enhanced statistical visualization and clearer chart aesthetics.

➤ **Plotly Express**

Used to generate interactive geographic maps such as choropleth maps.

Geospatial Libraries

➤ **GeoPandas**

Provides geospatial data handling capabilities.

➤ **Folium**

Used for interactive map visualizations if required.

4. Data Preparation and Cleaning

Before performing geographic analysis, the dataset was prepared and cleaned.

Date Conversion

The declaration date column was converted into datetime format.

```
df['declaration_date'] = pd.to_datetime(df['declaration_date'], errors='coerce')
```

This conversion allows extraction of time components such as year and month.

Feature Extraction

Two new columns were created:

```
df['year'] = df['declaration_date'].dt.year
```

```
df['month'] = df['declaration_date'].dt.month
```

These columns allow disaster data to be analyzed by time.

Column Renaming

```
df = df.rename(columns={"incident_type": "incidentType"})
```

The column name was simplified to make it easier to reference during analysis.

5. Geographic Aggregation

To perform spatial analysis, disaster data was aggregated by location.

Aggregation groups multiple disaster records into summarized counts.

5.1 State-Level Disaster Aggregation

Disaster declarations were grouped by state and counted.

```
state_counts = df.groupby('state').size().reset_index(name='disaster_count')
```

Method Explanation

The groupby() function groups records by the state column.

The size() function counts how many disaster declarations exist for each state.

The result is a dataset containing:

```
| State | Disaster Count |
```

This aggregation helps identify which states experience disasters most frequently.

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6. Geographic Visualization

After aggregation, several visualizations were created to reveal spatial patterns in disaster occurrences.

6.1 Choropleth Map – Disaster Hotspots

A choropleth map was created to visualize disaster frequency across US states.

➤ Method

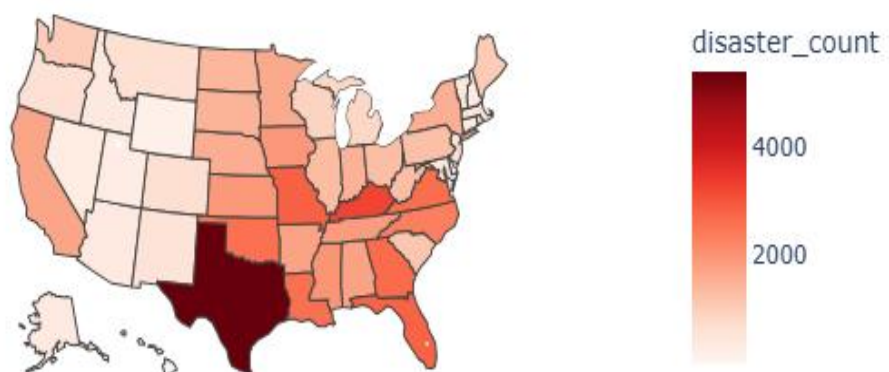
Plotly Express was used to generate the map.

px.choropleth()

➤ Key Parameters

Parameter	Purpose
locations	Specifies state identifiers
locationmode	Indicates US state codes
color	Represents disaster frequency
scope	Restricts the map to the United States
color_continuous_scale	Defines color gradient

US Disaster Declarations by State



➤ Interpretation

The map highlights disaster hotspots across the United States.

States with darker shades represent higher numbers of disaster declarations.

From the visualization it is evident that states such as Texas, California, and Florida experience significantly higher disaster frequencies.

This may be due to geographic and climatic factors such as:

- hurricanes along coastal regions
- wildfires in western states
- floods and storms in southern state

6.2 Top 10 Disaster-Prone States

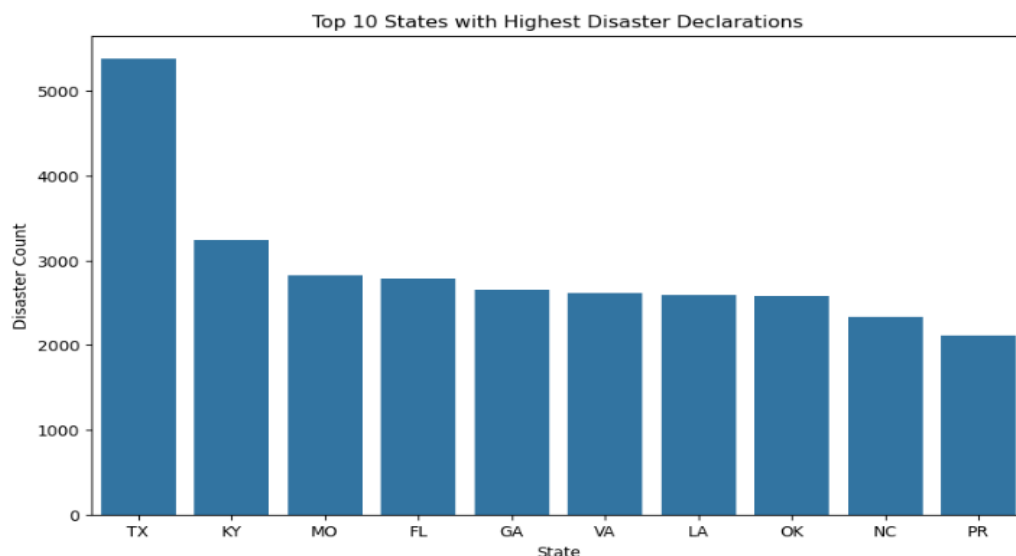
To compare disaster activity across states, a bar chart was created showing the top 10 states with the highest disaster declarations.

➤ Method

The dataset was sorted in descending order of disaster counts.

```
top_states = state_counts.sort_values(by='disaster_count', ascending=False).head(10)
```

A bar chart was generated using Seaborn.



➤ Interpretation

The bar chart clearly highlights the most disaster-prone states.

Texas appears as the state with the highest number of disaster declarations, followed by states such as California and Florida.

These states are exposed to multiple disaster risks including hurricanes, floods, and wildfires.

6.3 Incident Type Distribution by State

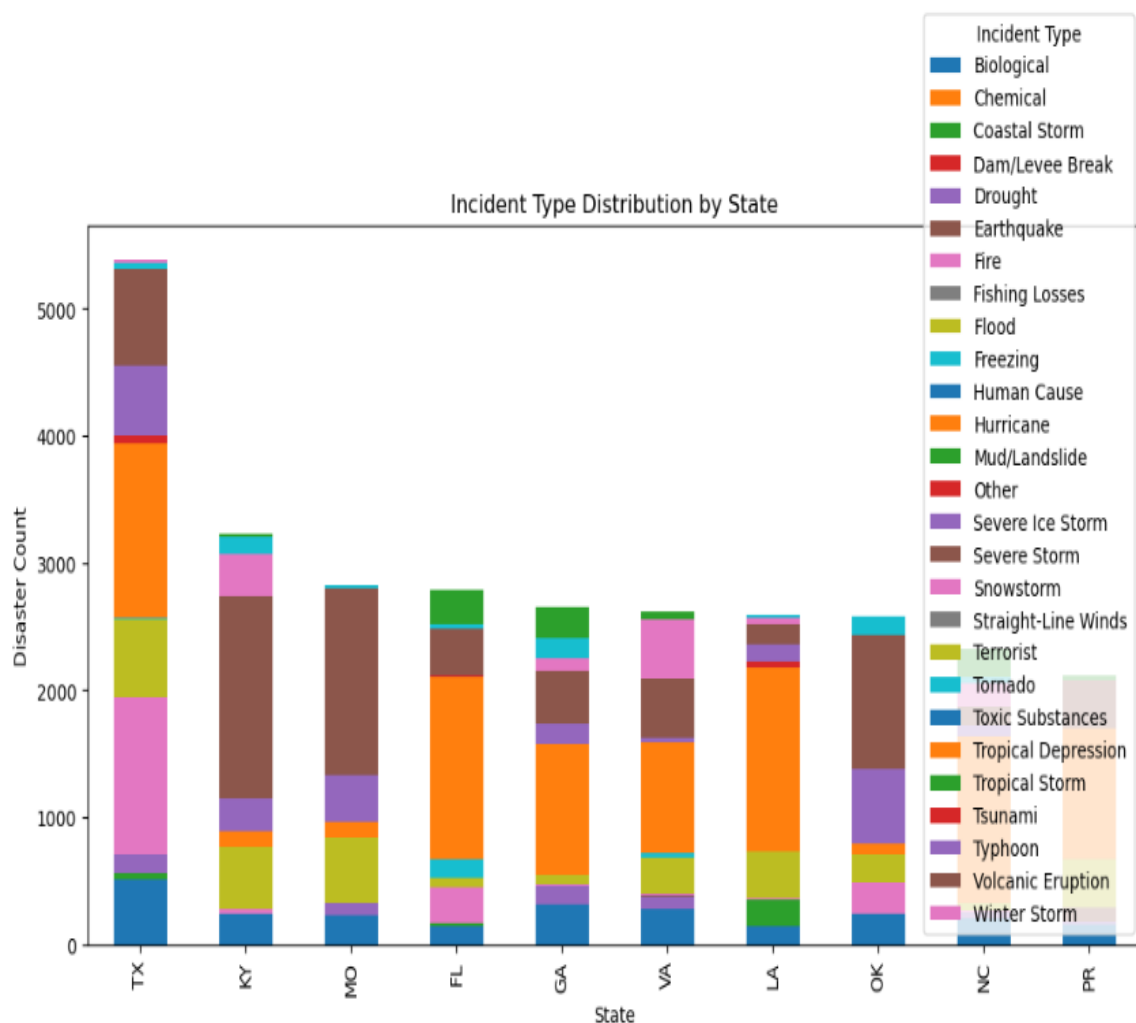
To understand which disaster types occur in different states, the dataset was grouped by state and disaster type.

```
incident_state = df.groupby(['state','incidentType']).size().reset_index(name='count')
```

A pivot table was created to restructure the dataset.

```
pivot_data = incident_state.pivot(index='state', columns='incidentType', values='count')
```

This allowed the creation of a stacked bar chart.



➤ Interpretation

The stacked bar chart shows how different disaster types contribute to the total disaster count in each state.

Some states show higher proportions of floods and storms, while others show higher wildfire or hurricane occurrences.

This visualization helps identify dominant disaster types for different regions.

6.4 Disaster Type Hotspot Map

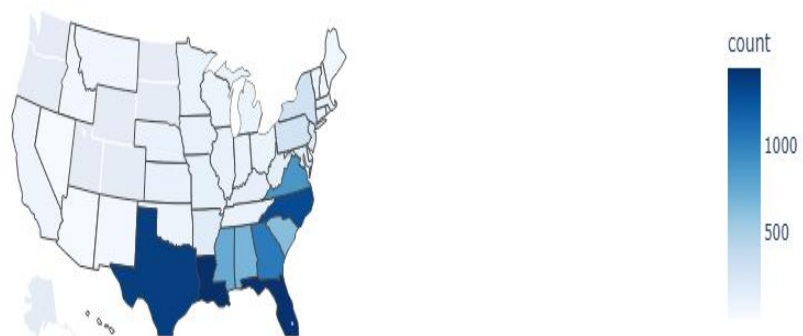
To further explore spatial patterns, a hotspot map was created for hurricanes.

The dataset was filtered for hurricane events.

```
hurricane_df = df[df['incidentType'] == 'Hurricane']
```

A choropleth map was generated showing hurricane occurrences by state.

Hurricane Hotspots in the US



➤ Interpretation

The hurricane hotspot map shows that coastal states such as Florida, Texas, and Louisiana have the highest hurricane frequencies.

This pattern is expected because these states are located along the Gulf of Mexico and Atlantic Ocean where tropical storms frequently form.

7. Key Insights from Geographic Analysis

The geographic analysis revealed several important patterns.

1. Disaster declarations are not evenly distributed across the United States.
2. Southern and coastal states experience higher disaster frequencies.
3. Different regions experience different types of disasters.
4. Coastal states are particularly vulnerable to hurricanes and tropical storms.
5. Western states often experience wildfire-related disasters.

These findings highlight how geography strongly influences disaster patterns.

8. Conclusion

In this milestone, geographic analysis of disaster declarations was performed to understand the spatial distribution of disasters across the United States.

The dataset was cleaned, aggregated, and visualized using multiple geographic visualization techniques including choropleth maps, bar charts, and stacked bar charts.

The analysis identified several disaster hotspots and revealed how disaster types vary across states.

These insights help improve understanding of disaster risk patterns and can support future disaster preparedness and mitigation strategies.